

Finite-Scenario Skew- t Regime Models for Trading, Risk-Neutralization, and Derivative Pricing

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Scope and caution. This study develops a finite-scenario modelling framework for research, model development, and controlled pricing experiments. It is not investment advice, a claim of unique arbitrage-free completion of an incomplete market, or a substitute for independent model validation. The central construction deliberately treats the empirical finite-support distribution as the operative model. It must not be represented as a consistent estimator of a nonexistent population MGF for an unbounded Student- t or skew- t law.

Abstract

This revised study (v2) treats the finite scenario law, rather than the empirical MGF, as the primitive pricing object (originally formulated that way in v1 of this document). The empirical MGF is an exact transform of that governed finite-support law and does not repair the nonexistent population MGF of an unbounded Student- t or skew- t distribution.

This study develops a unified architecture in which physical returns are modelled by a Markov-switching mixture of standardized skewed Student- t distributions, while derivative pricing is performed using finite empirical scenario laws and their exact empirical moment-generating functions. The skew parameter nests the symmetric Student- t model, and large degrees of freedom approach the Gaussian limit. However, an unbounded Student- t or skew- t log-return law has no finite moment-generating function at any nonzero real argument. In particular, setting the degrees of freedom to 30 does not make $E[e^X]$ finite.

The proposed resolution is not to claim that the population MGF exists. Instead, the physical model is discretized into a finite probability law. Its empirical MGF is finite by construction and is the exact MGF of that discrete model. Risk-neutral probabilities can then be obtained by empirical Esscher tilting, minimum-relative-entropy projection, or option-constrained entropy calibration. Regime transitions and emissions are jointly constrained so that the discounted underlying is a martingale under the selected pricing measure.

The framework supports direct finite-sum valuation, deterministic convolution, Monte Carlo, Fourier quadrature, Carr-Madan FFT, COS-type methods, and dynamic programming. The study derives the relevant transforms, identifies feasibility and non-uniqueness conditions, distinguishes a log-return Esscher tilt from an entropy projection constrained on gross returns, and specifies tail-completion, numerical-stability, no-arbitrage, convergence, and model-risk controls. It also proposes trading signals based on physical probabilities and relative-value signals based on differences between physical and risk-neutral distributions.

1. Executive conclusions

1. A mixed skew- t regime model is a coherent physical return model:

$$X_{t+1} = \mu_{S_{t+1}}^P + \sigma_{S_{t+1}}^P Z_{t+1}, \quad Z_{t+1} \mid S_{t+1} = k \sim \text{SkewT}^*(\nu_k^P, \lambda_k^P).$$

2. In a standardized Hansen-style parameterization, $\lambda_k = 0$ returns the symmetric Student- t . As $\nu_k \rightarrow \infty$, the symmetric Student- t converges in distribution to a Gaussian. A t_{30} law is close to Gaussian in its central body but still has polynomial tails and no nontrivial MGF [1, 2].

3. For any finite ν , a genuine unbounded Student- t log return satisfies

$$E[e^{uX}] = \infty$$

for every real $u \neq 0$. The same problem remains for common skew- t constructions with polynomial tails on both sides. Therefore, a raw skew- t log-price model cannot directly satisfy the stock martingale condition through a population MGF.

4. Given finite scenario nodes x_i and positive probabilities p_i , the empirical law

$$\widehat{F}(dx) = \sum_{i=1}^N p_i \delta_{x_i}(dx)$$

has the exact empirical MGF

$$\widehat{M}(u) = \sum_{i=1}^N p_i e^{ux_i},$$

which is finite for all finite real or complex u .

5. This finiteness is a property of the chosen finite-support model. It is not evidence that the empirical MGF converges to a finite population MGF as $N \rightarrow \infty$ under an unbounded Student- t data-generating law.

6. An empirical Esscher tilt is valid on the finite scenario space:

$$q_i(\theta) = \frac{p_i e^{\theta x_i}}{\sum_j p_j e^{\theta x_j}}.$$

Under this tilt,

$$\widehat{\kappa}_Q(u; \theta) = \widehat{\kappa}_P(u + \theta) - \widehat{\kappa}_P(\theta).$$

7. A minimum-relative-entropy projection subject directly to the tradable gross-return constraint is generally **not** the same as a log-return Esscher tilt. If the constraint is $E_Q[e^X] = F/S_0$, the entropy solution has the form

$$q_i(\beta) = \frac{p_i \exp(\beta e^{x_i})}{\sum_j p_j \exp(\beta e^{x_j})},$$

where the sign of β depends only on the Lagrange-multiplier convention. More generally, constraints on payoff features $g_m(x)$ produce an exponential family in those features.

8. In a regime-switching model, no-arbitrage constrains the combination of risk-neutral transition probabilities and regime emission laws. It does not generally determine them uniquely. Option prices or an explicit selection principle are required to identify the remaining risk premia [3, 4].

9. A production framework should use the empirical transform as a computational device, while explicitly governing support, tail completion, maturity consistency, weight concentration, and option-surface arbitrage.

2. Mathematical preliminaries

2.1 Prices, returns, and measures

Let S_t be the ex-dividend spot price. Over one model step of length Δ , define the log return

$$X_{t+\Delta} = \log \left(\frac{S_{t+\Delta}}{S_t} \right).$$

Let P denote the physical measure used for forecasting and trading analysis, and let Q denote a selected equivalent martingale measure used for pricing. With deterministic short rate r_t and dividend yield q_t , define the one-step carry

$$c_t = \int_t^{t+\Delta} (r_u - q_u) du.$$

The one-step martingale restriction is

$$E_Q [e^{X_{t+\Delta}} | \mathcal{F}_t] = e^{c_t}.$$

Equivalently,

$$E_Q[S_{t+\Delta} | \mathcal{F}_t] = S_t e^{c_t}.$$

This condition concerns the exponential of the log return, not merely the mean of the log return.

2.2 MGF, CGF, and characteristic function

For a random variable X , the MGF is

$$M_X(u) = E[e^{uX}],$$

where the expectation is finite. The cumulant-generating function is

$$\kappa_X(u) = \log M_X(u).$$

The characteristic function is

$$\phi_X(v) = E[e^{ivX}], \quad v \in \mathbb{R}.$$

The characteristic function always exists because $|e^{ivX}| = 1$. The MGF need not exist because e^{uX} is unbounded for real $u \neq 0$.

2.3 Why Student- t has no nontrivial MGF

For a Student- t variable with ν degrees of freedom, the density has polynomial tails:

$$f_X(x) \asymp C|x|^{-(\nu+1)}$$

as $|x| \rightarrow \infty$. For any $u > 0$, the positive tail of

$$\int_{-\infty}^{\infty} e^{ux} f_X(x) dx$$

behaves like $e^{ux} x^{-(\nu+1)}$ and diverges. For any $u < 0$, the negative tail diverges. Hence

$$M_X(u) = \infty, \quad u \neq 0.$$

The existence of many ordinary moments when ν is large does not imply existence of the MGF. A Student- t with $\nu = 30$ has finite moments only up to orders below 30, while an MGF finite in a neighborhood of zero would imply moments of all orders.

3. Physical mixed skew- t regime model

3.1 Hidden regime process

Let

$$S_t \in \{1, \dots, K\}$$

be a latent regime following a first-order Markov chain:

$$p_{ij}^P = P(S_{t+1} = j \mid S_t = i).$$

The matrix $P^P = (p_{ij}^P)$ is row stochastic:

$$p_{ij}^P \geq 0, \quad \sum_{j=1}^K p_{ij}^P = 1.$$

Hamilton's filtering framework provides a standard basis for estimating latent discrete-state models and forecasting regime probabilities [5].

3.2 Standardized skew- t emissions

Use

$$X_{t+1} = \mu_{S_{t+1}}^P + \sigma_{S_{t+1}}^P Z_{t+1},$$

where

$$Z_{t+1} \mid S_{t+1} = k \sim \text{SkewT}^*(\nu_k^P, \lambda_k^P),$$

and the standardized innovation satisfies

$$E_P[Z_{t+1} \mid S_{t+1} = k] = 0, \quad \text{Var}_P(Z_{t+1} \mid S_{t+1} = k) = 1.$$

Accordingly,

$$E_P[X_{t+1} \mid S_{t+1} = k] = \mu_k^P,$$

and

$$\text{Var}_P(X_{t+1} \mid S_{t+1} = k) = (\sigma_k^P)^2.$$

This standardization is operationally important because in several skew- t parameterizations the raw location parameter is not the mean and the raw scale parameter is not the standard deviation.

3.3 Nesting structure

For the Hansen skewed Student density [1]:

$$\lambda_k = 0 \implies \text{SkewT}^*(\nu_k, \lambda_k) = t_{\nu_k}^*.$$

Then

$$\nu_k \rightarrow \infty \implies t_{\nu_k}^* \Rightarrow N(0, 1).$$

Thus the parameter space contains a symmetric Student- t submodel and approaches a Gaussian limit.

For the symmetric t_ν with $\nu > 4$, excess kurtosis is

$$\gamma_2 = \frac{6}{\nu - 4}.$$

At $\nu = 30$,

$$\gamma_2 = \frac{6}{26} \approx 0.230769.$$

Therefore t_{30} is visibly close to normal in the body but is not exactly Gaussian and remains fundamentally different in the far tails.

3.4 Suggested parsimonious starting model

A three-state model may be initialized as follows:

- Quiet state: $\lambda_1 = 0, \nu_1 = 30$.
- Stress state: $\lambda_2 < 0, 4 < \nu_2 < 12$.
- Recovery or squeeze state: $\lambda_3 > 0, 4 < \nu_3 < 15$.

A stable first specification can constrain

$$\nu_2 = \nu_3 = \nu_{\text{tail}}$$

and optionally

$$\lambda_3 = -\lambda_2.$$

These restrictions can later be relaxed only if the resulting parameters remain identifiable and improve out-of-sample density and tail forecasts.

3.5 Physical filtering and forecasting

Let

$$\alpha_{t,k}^P = P(S_t = k \mid \mathcal{F}_t)$$

be the filtered state probabilities. Prediction gives

$$\alpha_{t+1|t,j}^P = \sum_{i=1}^K \alpha_{t,i}^P p_{ij}^P.$$

After observing $X_{t+1} = x$, update by Bayes' rule:

$$\alpha_{t+1,j}^P = \frac{\alpha_{t+1|t,j}^P f_j^P(x)}{\sum_{\ell=1}^K \alpha_{t+1|t,\ell}^P f_\ell^P(x)}.$$

For live trading and historical backtesting, filtered probabilities must use only information available at the relevant time. Smoothed probabilities condition on future observations and therefore must not be used as if they were contemporaneously observable signals.

4. Finite empirical scenario law

4.1 Definition

For regime k , define scenario nodes and physical weights

$$\{(x_{k,i}, p_{k,i}^P)\}_{i=1}^{N_k},$$

with

$$p_{k,i}^P > 0, \quad \sum_{i=1}^{N_k} p_{k,i}^P = 1.$$

The empirical regime law is

$$\widehat{F}_k^P(dx) = \sum_{i=1}^{N_k} p_{k,i}^P \delta_{x_{k,i}}(dx).$$

The strict positivity requirement is convenient if equivalence of P and Q on the finite support is desired. If $p_i^P = 0$, an absolutely continuous change of measure cannot assign that node positive Q -probability.

4.2 Construction alternatives

Weighted historical law

If $\gamma_{t,k}$ is a regime responsibility, define

$$p_{k,t}^P = \frac{\gamma_{t,k}}{\sum_s \gamma_{s,k}}.$$

This uses observed returns but may undersample rare tails.

Deterministic quantile grid

Choose probabilities u_i and define

$$x_{k,i} = F_k^{-1}(u_i).$$

For an equal-probability midpoint grid,

$$u_i = \frac{i - \frac{1}{2}}{N_k}, \quad p_{k,i}^P = \frac{1}{N_k}.$$

A tail-enriched grid uses nonuniform cells and assigns each node the exact probability mass of its cell.

Hybrid body and stress nodes

Combine historical observations, fitted skew- t quantiles, and explicit stress nodes. The probabilities must then be normalized and documented. Stress-node probabilities are model assumptions, not empirical facts.

4.3 Empirical MGF and CGF

For regime k ,

$$\widehat{M}_k^P(u) = \sum_{i=1}^{N_k} p_{k,i}^P e^{ux_{k,i}},$$

and

$$\widehat{\kappa}_k^P(u) = \log \widehat{M}_k^P(u).$$

Because the support is finite, $\widehat{M}_k^P(u)$ is finite for every finite complex u .

The derivatives satisfy

$$\frac{d}{du} \widehat{\kappa}_k^P(u) = \sum_i w_{k,i}(u) x_{k,i},$$

where

$$w_{k,i}(u) = \frac{p_{k,i}^P e^{ux_{k,i}}}{\sum_j p_{k,j}^P e^{ux_{k,j}}},$$

and

$$\frac{d^2}{du^2} \widehat{\kappa}_k^P(u) = \text{Var}_{w_k(u)}(X_k) \geq 0.$$

Thus the empirical CGF is convex and strictly convex unless all active nodes are equal.

4.4 Stable numerical evaluation

Define

$$a_i(u) = \log p_i + ux_i, \quad m(u) = \max_i a_i(u).$$

Then

$$\widehat{\kappa}(u) = m(u) + \log \sum_i \exp(a_i(u) - m(u)).$$

This log-sum-exp form should be mandatory. Direct exponentiation can overflow or underflow and may silently corrupt the risk-neutral calibration.

5. What empirical-MGF finiteness does and does not mean

5.1 Exact statement

For fixed finite nodes and weights, $\widehat{M}_N(u)$ is the exact MGF of the empirical discrete law. Every exponential moment exists.

5.2 Incorrect statement to avoid

It is incorrect to claim that

$$\widehat{M}_N(u) \rightarrow M_X(u) < \infty$$

for samples drawn from an unbounded Student- t population and real $u \neq 0$.

5.3 Extreme-node instability

Let

$$X_{(N)} = \max_{1 \leq i \leq N} X_i.$$

For equal weights,

$$\widehat{M}_N(1) = \frac{1}{N} \sum_{i=1}^N e^{X_i} \geq \frac{e^{X_{(N)}}}{N}.$$

As new heavy-tailed observations increase the maximum, the empirical exponential moment can jump discontinuously in economic terms. Apparent stability over a finite historical window is not convergence evidence.

5.4 Correct interpretation

The empirical MGF defines a **finite-horizon, finite-support model**. Its validity depends on:

- the support policy;
- tail-node design;
- scenario probabilities;
- calibration instruments;
- maturity horizon;
- sensitivity to extreme nodes.

6. Empirical risk-neutralization

6.1 Empirical log-return Esscher tilt

Define

$$q_i^E(\theta) = \frac{p_i^P e^{\theta x_i}}{\sum_j p_j^P e^{\theta x_j}}.$$

Then

$$\widehat{M}_Q^E(u; \theta) = \sum_i q_i^E(\theta) e^{ux_i} = \frac{\widehat{M}_P(u + \theta)}{\widehat{M}_P(\theta)},$$

and

$$\widehat{\kappa}_Q^E(u; \theta) = \widehat{\kappa}_P(u + \theta) - \widehat{\kappa}_P(\theta).$$

This is the exact Esscher identity on the empirical support. Esscher-based option pricing selects a martingale measure by exponential tilting when the relevant transform exists [3].

6.2 One-regime martingale equation

For one period with carry c , require

$$\widehat{M}_Q^E(1; \theta) = e^c.$$

Equivalently,

$$\boxed{\widehat{\kappa}_P(\theta + 1) - \widehat{\kappa}_P(\theta) = c.}$$

Define

$$g(\theta) = \widehat{\kappa}_P(\theta + 1) - \widehat{\kappa}_P(\theta).$$

The derivative is

$$g'(\theta) = \widehat{\kappa}'_P(\theta + 1) - \widehat{\kappa}'_P(\theta) \geq 0$$

because $\widehat{\kappa}'_P$ is nondecreasing. With a nondegenerate support, strict monotonicity normally holds.

As $\theta \rightarrow -\infty$, the tilted mass concentrates on $x_{\min}$, giving

$$g(\theta) \rightarrow x_{\min}.$$

As $\theta \rightarrow +\infty$,

$$g(\theta) \rightarrow x_{\max}.$$

Therefore a finite solution exists when

$$x_{\min} < c < x_{\max}.$$

At the boundary, the solution requires a limiting degenerate measure concentrated at an extreme node, not a finite θ . For a tradable gross return, this condition is equivalently

$$e^{x_{\min}} < e^c < e^{x_{\max}}.$$

6.3 Minimum-relative-entropy projection

Instead of restricting Q to a log-return Esscher family, solve

$$\min_{q \in \Delta_N} D_{\text{KL}}(q \| p) = \min_q \sum_i q_i \log \left(\frac{q_i}{p_i} \right)$$

subject to

$$\sum_i q_i = 1, \quad \sum_i q_i g_m(x_i) = b_m, \quad m = 1, \dots, M.$$

The solution has exponential-family form

$$q_i(\beta) = \frac{p_i \exp \left(\sum_{m=1}^M \beta_m g_m(x_i) \right)}{\sum_j p_j \exp \left(\sum_{m=1}^M \beta_m g_m(x_j) \right)}.$$

This follows from the Lagrangian first-order conditions. Maximum-entropy and relative-entropy projections under expectation constraints generate exponential-family solutions [6, 7].

If the sole pricing constraint is the forward condition,

$$g_1(x) = e^x, \quad b_1 = e^c,$$

then

$$q_i(\beta) = \frac{p_i \exp(\beta e^{x_i})}{\sum_j p_j \exp(\beta e^{x_j})}.$$

This is generally different from

$$q_i^E(\theta) \propto p_i e^{\theta x_i}.$$

The two coincide only in special or approximate circumstances. Confusing these two tilts is a material mathematical error.

6.4 Feasibility of entropy calibration

The constraint vector $\mathbf{b} = (b_1, \dots, b_M)$ must lie in the convex hull of the feature vectors

$$\mathbf{g}(x_i) = (g_1(x_i), \dots, g_M(x_i)).$$

For the one-dimensional forward constraint, feasibility requires

$$\min_i e^{x_i} \leq e^c \leq \max_i e^{x_i}.$$

Strict interior feasibility yields strictly positive calibrated probabilities. Boundary feasibility generally produces concentration on boundary nodes.

6.5 Option-constrained entropy calibration

At maturity T , let terminal node i imply $S_{T,i}$ and discounted payoff

$$G_{m,i} = D(0, T) \Pi_m(S_{T,i}).$$

A robust calibration is

$$\min_q D_{\text{KL}}(q \| p) + \lambda_R \mathcal{R}(q)$$

subject to

$$q_i \geq 0, \quad \sum_i q_i = 1,$$

$$\sum_i q_i S_{T,i} = F_{0,T},$$

and bid-ask constraints

$$V_m^{\text{bid}} \leq \sum_i q_i G_{m,i} \leq V_m^{\text{ask}}.$$

Exact equality to option mids should be used only when quotes are reliable and jointly arbitrage-consistent. Option prices reveal state-price information through strike derivatives, forming the basis of risk-neutral density recovery [8].

6.6 Incompleteness and selection of Q

A martingale condition supplies restrictions, not a unique risk-neutral measure, when the model contains unhedgeable regime and jump risks. Empirical Esscher tilting, minimum entropy, and direct option calibration are alternative measure-selection rules. They should be compared rather than treated as universally equivalent [3, 4].

7. Regime-switching risk-neutral dynamics

7.1 Risk-neutral transitions

Let

$$P^Q = (p_{ij}^Q)$$

be the risk-neutral transition matrix. In general,

$$P^Q \neq P^P.$$

The difference represents regime-transition risk premia and must be inferred from option prices, constrained economically, or selected through a pricing-kernel principle.

7.2 Emissions attached to the arriving state

Assume the one-step return distribution depends on $S_{t+1} = j$. Let

$$m_j^Q(u) = E_Q[e^{uX_{t+1}} \mid S_{t+1} = j]$$

be the regime empirical MGF. Define

$$D^Q(u) = \text{diag}(m_1^Q(u), \dots, m_K^Q(u)).$$

With current state-probability column vector α_t^Q , the h -step MGF is

$$M_{t,h}^Q(u) = (\alpha_t^Q)^\top [P^Q D^Q(u)]^h \mathbf{1}.$$

This ordering is correct for emissions associated with the **arriving** state. If emissions are associated with the departing state, the ordering changes to $D^Q(u)P^Q$. The convention must be fixed consistently throughout estimation, simulation, and pricing.

The characteristic function is

$$\phi_{t,h}^Q(v) = M_{t,h}^Q(iv).$$

7.3 Conditional martingale restriction by current state

For each current state i ,

$$E_Q[e^{X_{t+1}} \mid S_t = i] = \sum_{j=1}^K p_{ij}^Q m_j^Q(1).$$

The one-step stock martingale condition requires

$$\sum_{j=1}^K p_{ij}^Q m_j^Q(1) = e^{c_t}, \quad i = 1, \dots, K.$$

In matrix form,

$$P^Q \mathbf{m}^Q(1) = e^{c_t} \mathbf{1}.$$

A sufficient but stronger restriction is

$$m_j^Q(1) = e^{c_t}$$

for every regime j . This makes the martingale condition hold for any transition matrix, but it suppresses possible compensation between transition risk and state-conditional expected gross returns. It is a modelling choice, not a necessity.

7.4 Multi-step martingale consistency

If the one-step condition holds conditionally for every current state and carry is deterministic, iterated expectations imply

$$E_Q[e^{X_{t,h}} \mid \mathcal{F}_t] = e^{\sum_{\ell=0}^{h-1} c_{t+\ell}},$$

where

$$X_{t,h} = \sum_{\ell=1}^h X_{t+\ell}.$$

If rates, dividends, or transition matrices are state dependent, the discounting and state process must be included jointly. A scalar deterministic correction is then generally insufficient.

8. Trading models under the physical measure

8.1 Horizon distribution

The physical h -step characteristic function can be propagated as

$$\phi_{t,h}^P(v) = (\alpha_t^P)^\top [P^P D^P(iv)]^h \mathbf{1}.$$

Alternatively, the full distribution can be generated by deterministic convolution or Monte Carlo.

8.2 Trading quantities

Expected log return:

$$\mu_{t,h}^P = E_P[X_{t,h} \mid \mathcal{F}_t].$$

Probability of a positive return:

$$p_{t,h}^+ = P_P(X_{t,h} > 0 \mid \mathcal{F}_t).$$

Target-hit probability:

$$p_{t,h}^G = P_P(X_{t,h} > G \mid \mathcal{F}_t).$$

Stop-loss probability:

$$p_{t,h}^L = P_P(X_{t,h} < -L \mid \mathcal{F}_t).$$

Physical Value at Risk for loss $L_{t,h} = -\Pi_{t,h}$:

$$\text{VaR}_\alpha^P = \inf\{\ell : P_P(L_{t,h} \leq \ell) \geq \alpha\}.$$

Expected Shortfall:

$$\text{ES}_\alpha^P = E_P[L_{t,h} \mid L_{t,h} \geq \text{VaR}_\alpha^P]$$

for continuous losses, with the usual quantile-integral definition preferred in discrete laws to handle atoms correctly:

$$\text{ES}_\alpha^P = \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_u^P du.$$

8.3 Trade-specific expected utility or payoff

For a proposed trade payoff $\Pi_{\text{trade}}(X_{t,h})$,

$$\text{Edge}_{t,h} = E_P[\Pi_{\text{trade}}(X_{t,h}) \mid \mathcal{F}_t] - \text{costs}.$$

A risk-adjusted score may be

$$\text{Score}_{t,h} = \frac{E_P[\Pi_{\text{trade}}] - \text{costs}}{\text{ES}_\alpha^P(L_{\text{trade}})}.$$

This is a design choice rather than a theorem. Position sizing must also include turnover, liquidity, limits, drawdown, and model uncertainty.

8.4 Physical versus risk-neutral divergence

For a region A ,

$$\Delta p_{t,h}(A) = P_P(X_{t,h} \in A \mid \mathcal{F}_t) - P_Q(X_{t,h} \in A \mid \mathcal{F}_t).$$

This can support option relative-value analysis, but the difference includes risk premia, market frictions, and model error. It is not automatically an arbitrage signal.

9. Derivative pricing engines

9.1 Direct finite-sum valuation

For a one-period payoff $\Pi(S_T)$ under terminal nodes x_i ,

$$V_0 = D(0, T) \sum_i q_i^Q \Pi(S_0 e^{x_i}).$$

This is exact under the finite empirical model. It is the primary unit-test benchmark for other engines.

9.2 Deterministic convolution

For independent conditional increments, finite discrete laws can be convolved. In a regime model, propagate a vector of state-conditioned mass functions. After each step:

1. transition regime probabilities;
2. convolve with the arriving-state emission law;
3. project onto a controlled log-price grid if necessary;
4. preserve total mass and the discounted first moment.

Projection introduces numerical error and must be separated from model error.

9.3 Monte Carlo

For each risk-neutral path:

1. draw the next state from P^Q ;
2. draw a scenario node from the corresponding $q_{k,i}^Q$;
3. update $S_{t+\Delta} = S_t e^{x_{k,i}}$;
4. continue until maturity;
5. evaluate and discount the payoff.

The estimator is

$$\widehat{V}_0^{MC} = D(0, T) \frac{1}{M} \sum_{m=1}^M \Pi^{(m)}.$$

Its estimated standard error is

$$\widehat{\text{SE}}(\widehat{V}_0^{MC}) = D(0, T) \frac{s_\Pi}{\sqrt{M}}.$$

Monte Carlo is appropriate for barriers, Asians, cliquets, callable structures, and other path-dependent products. Variance reduction should use controls such as a matched Gaussian model, stratification by state path, antithetics where meaningful, conditional expectation, or quasi-Monte Carlo.

9.4 Fourier transform and FFT

The empirical terminal characteristic function is

$$\phi_T^Q(v) = \sum_i q_i^Q e^{ivx_i}.$$

For the regime model it is obtained through the matrix recursion above.

Carr-Madan pricing applies exponential damping to a call-price function and evaluates the characteristic function at a complex-shifted argument [9]. For damping parameter $\alpha > 0$, sufficient integrability includes

$$E_Q[S_T^{\alpha+1}] < \infty.$$

In the empirical finite-support model,

$$E_Q[e^{(\alpha+1)X_T}] < \infty$$

for every finite α , but only because the support is finite. The admissible numerical α should still be chosen to control overflow, truncation, and oscillation.

The shifted transform can be evaluated directly as

$$\phi_T^Q(v - i(\alpha + 1)) = E_Q[e^{ivX_T} e^{(\alpha+1)X_T}].$$

FFT is valuable for pricing many strikes simultaneously, while direct Fourier quadrature is often simpler and more accurate for isolated strikes. Carr and Madan show how a known characteristic function can be used for efficient option valuation by FFT [9].

9.5 Quadrature

For a smoothed terminal density $f_T^Q(x)$,

$$V_0 = D(0, T) \int_{-\infty}^{\infty} \Pi(S_0 e^x) f_T^Q(x) dx.$$

For an unsmoothed empirical law, this integral is exactly a finite sum. Quadrature becomes relevant after kernel smoothing, interpolated density construction, or continuous tail completion.

Recommended methods include:

- adaptive Gauss-Kronrod;
- domain splitting at payoff kinks;
- double-exponential quadrature for difficult tails;
- Gauss-Legendre after a finite-domain map;
- Gauss-Hermite when a conditional Gaussian representation is used.

9.6 American and path-dependent products

American pricing requires dynamic programming or least-squares Monte Carlo. The state vector should include at least spot and latent-state information. If only filtered regime probabilities are observable, exercise policies should be expressed in terms of (S_t, α_t^Q) rather than an unobservable true state.

10. Greeks and hedging

10.1 Delta under a fixed terminal empirical law

For a European call,

$$C_0 = D(0, T) \sum_i q_i^Q (S_0 e^{x_i} - K)^+.$$

If the calibrated weights and return nodes are held fixed with respect to S_0 , then away from node-specific payoff kinks,

$$\Delta = D(0, T) \sum_i q_i^Q e^{x_i} \mathbf{1}_{\{S_0 e^{x_i} > K\}}.$$

10.2 Calibration-aware Greeks

In production, q_i^Q , nodes, forwards, and implied parameters may depend on spot. A full Greek must account for recalibration or specify a sticky rule. The fixed-law derivative above is not automatically the desk hedge.

10.3 Gamma roughness

A purely discrete terminal law produces atoms and therefore non-smooth strike and spot Greeks. Remedies include:

- kernel smoothing;
- entropy-smoothed interpolation;
- local continuous distributions around nodes;
- option-consistent spline densities;
- a sufficiently fine convolution grid.

Smoothing must preserve probability, the forward, and nonnegative density.

11. Tail policy

11.1 Why tail policy cannot be omitted

The empirical model cannot generate values outside its support. Deep-out-of-the-money options and long-horizon exponential moments can therefore be dominated by arbitrary endpoint choices.

11.2 Permissible policies

Hard bounded support

Specify

$$L_k \leq X_k \leq U_k.$$

This is transparent but imposes absolute bounds.

Truncated skew- t

For fitted skew- t CDF F_k , choose

$$L_k = F_k^{-1}(\varepsilon_k^-), \quad U_k = F_k^{-1}(1 - \varepsilon_k^+).$$

The truncated density is

$$f_k^{\text{tr}}(x) = \frac{f_k(x) \mathbf{1}_{[L_k, U_k]}(x)}{F_k(U_k) - F_k(L_k)}.$$

Continuous tail completion

Use an empirical or skew- t body and attach integrable left and right tails. This is effectively a new tempered or spliced distribution and must be labelled accordingly.

Option-implied tail nodes

Add nodes beyond historical support and infer their Q weights from liquid option prices under regularization. This is the preferred empirical derivative-pricing approach when strikes extend beyond the historical scenario range.

11.3 Boundary sensitivity

Every material output should be recalculated under several support policies. A price that changes substantially when a single remote node is added is tail-policy dominated.

12. Calibration architecture

12.1 Physical calibration

Estimate

$$\Theta^P = \{P^P, \mu_k^P, \sigma_k^P, \nu_k^P, \lambda_k^P\}_{k=1}^K$$

by maximum likelihood, Bayesian inference, or penalized likelihood.

Useful constraints include

$$\sigma_k^P > 0, \quad \nu_k^P > 2, \quad -1 < \lambda_k^P < 1$$

for the Hansen parameterization. If fourth moments are operationally needed, constrain

$$\nu_k^P > 4.$$

Use state-ordering restrictions, such as

$$\sigma_1^P < \sigma_2^P < \dots < \sigma_K^P,$$

to reduce label switching.

12.2 Risk-neutral calibration

Risk-neutral parameters may include

$$P^Q, \quad \theta_k, \quad q_{k,i,T}^Q,$$

subject to martingale and arbitrage constraints.

A representative objective is

$$\mathcal{J} = \sum_m w_m \left(\frac{V_m^{\text{model}} - V_m^{\text{mid}}}{\max(s_m, s_{\min})} \right)^2 + \lambda_{\text{KL}} D_{\text{KL}}(Q \| P) + \lambda_R \mathcal{R}(Q).$$

The regularizer may penalize:

- abrupt adjacent-node weight changes;
- transition matrices far from P^P ;
- concentrated weights;
- maturity-to-maturity instability;
- excessive tail-node mass.

12.3 Exact constraints versus penalties

Use exact constraints for:

- probability normalization;
- nonnegativity;
- the forward martingale condition;

- hard static-arbitrage conditions.

Use penalties or bid-ask inequalities for noisy option observations.

12.4 Maturity consistency

Calibrating a separate terminal distribution at each maturity does not guarantee that the family is generated by a common martingale process. A dynamic regime model, martingale transport condition, or explicit inter-maturity consistency layer is required.

13. No-arbitrage validation

For European options with deterministic rates and dividends:

13.1 Put-call parity

$$C(K, T) - P(K, T) = S_0 e^{-qT} - K e^{-rT}.$$

13.2 Bounds

$$\max(S_0 e^{-qT} - K e^{-rT}, 0) \leq C(K, T) \leq S_0 e^{-qT}.$$

13.3 Strike monotonicity

$$\frac{\partial C}{\partial K} \leq 0.$$

13.4 Strike convexity

$$\frac{\partial^2 C}{\partial K^2} \geq 0.$$

Under smoothness, the Breeden-Litzenberger identity gives

$$\frac{\partial^2 C}{\partial K^2} = D(0, T) f_{S_T}^Q(K),$$

linking option convexity to a nonnegative risk-neutral density [8]. For a discrete empirical law, the second derivative is a positive measure with atoms rather than an ordinary smooth density.

13.5 Forward martingale test

Numerically verify

$$\left| \sum_i q_i^Q S_{T,i} - F_{0,T} \right| < \varepsilon_F.$$

13.6 Calendar consistency

Check calendar-spread constraints in an appropriate forward-normalized representation. Static strike arbitrage at every maturity does not by itself guarantee dynamic consistency across maturities.

14. Numerical and statistical validation

14.1 Cross-engine checks

Use direct finite-sum or high-accuracy deterministic convolution as the benchmark. Require

$$|V_{\text{FFT}} - V_{\text{benchmark}}| < \varepsilon_{\text{FFT}},$$

and

$$|V_{\text{MC}} - V_{\text{benchmark}}| \leq z_{1-\alpha/2} \widehat{\text{SE}}_{\text{MC}} + \varepsilon_{\text{disc}}.$$

14.2 Scenario-count convergence

Test increasing N while holding the support policy fixed. Do not vary N and support simultaneously and then attribute all changes to discretization.

14.3 Support convergence

Vary tail probability cutoffs, endpoint nodes, and tail-completion methods separately.

14.4 Weight concentration

Effective scenario size:

$$N_{\text{eff}}^Q = \frac{1}{\sum_i (q_i^Q)^2}.$$

Entropy:

$$H(Q) = - \sum_i q_i^Q \log q_i^Q.$$

Maximum weight:

$$q_{\max}^Q = \max_i q_i^Q.$$

A very low N_{eff}^Q indicates that pricing is controlled by very few scenarios.

14.5 Leave-one-extreme-out analysis

Recalculate prices and risk measures after removing:

- the largest positive node;
- the most negative node;
- each major stress node;
- each sparse historical crisis episode.

14.6 Rolling stability

Monitor through time:

$$P^P, \quad P^Q, \quad \nu_k, \quad \lambda_k, \quad N_{\text{eff}}^Q, \quad \text{tail-node mass}.$$

14.7 Out-of-sample trading validation

Evaluate:

- predictive log score;
 - probability integral transform calibration;
 - left- and right-tail exceedance rates;
 - VaR conditional coverage;
 - ES backtests;
 - net P&L after costs;
 - turnover;
 - drawdown;
 - parameter stability.
-

15. Corrected conceptual points

15.1 LLN versus CLT

The law of large numbers concerns convergence of the sample mean to the population mean. The central limit theorem concerns the standardized sampling distribution of sums or means. Neither causes individual returns to become normal.

15.2 Symmetry versus reversal

A symmetric return density means equal centered left- and right-tail probabilities. It does not mean that a positive return must later be offset by a negative return of equal magnitude.

15.3 t_{30} versus Gaussian

t_{30} resembles a Gaussian over much of the body but retains polynomial tails. It therefore does not inherit the Gaussian MGF.

15.4 Empirical MGF versus population MGF

The empirical MGF is exact for the finite empirical law. It is not a regularization proof for the unbounded skew- t population.

15.5 Girsanov versus transforms

A change of measure specifies Q relative to P . An MGF or CGF is a transform used to characterize a chosen law and impose exponential-moment conditions. In Gaussian diffusion models, Girsanov and cumulant corrections are consistent pieces of the same risk-neutral construction, not interchangeable concepts.

15.6 Esscher tilt versus entropy projection

A log-return Esscher family uses the sufficient statistic x :

$$q_i \propto p_i e^{\theta x_i}.$$

A minimum-KL projection constrained on the gross return uses the sufficient statistic e^x :

$$q_i \propto p_i e^{\beta e^{x_i}}.$$

They are distinct unless the imposed feature is the same.

15.7 Regime-specific versus joint martingale constraints

Imposing $m_j^Q(1) = e^c$ in every state is sufficient but stronger than necessary. The correct condition for emissions attached to the arriving state is

$$P^Q \mathbf{m}^Q(1) = e^c \mathbf{1}.$$

15.8 Finite support and equivalence

Strictly positive P and Q weights on the same finite nodes give equivalent measures on that finite scenario space. This does not imply equivalence to a continuous unbounded skew- t law outside the chosen support.

16. Proposed production design

Layer 1: physical regime engine

Fit a standardized three-state skew- t HMM under P .

Layer 2: scenario engine

Construct deterministic quantile nodes, historical nodes, and governed stress nodes with documented probabilities.

Layer 3: empirical transform engine

Provide stable evaluations of

$$\widehat{M}_k(u), \quad \widehat{\kappa}_k(u), \quad \widehat{\phi}_k(v),$$

using log-sum-exp and complex-safe arithmetic.

Layer 4: risk-neutral calibration

Offer at least three selectable methods:

1. empirical log-return Esscher tilt;
2. minimum-KL projection under forward and payoff constraints;
3. full option-constrained entropy calibration.

Layer 5: horizon propagation

Use state-aware matrix transforms, deterministic convolution, or simulation based on product requirements.

Layer 6: pricing engines

Implement:

1. direct finite-sum benchmark;
2. deterministic grid convolution;

3. adaptive Fourier quadrature;
4. FFT or COS strip pricer;
5. Monte Carlo;
6. least-squares Monte Carlo for early exercise.

Layer 7: trading analytics

Produce physical regime probabilities, horizon return distributions, target probabilities, VaR, ES, expected payoff, and P -versus- Q divergence measures.

Layer 8: validation and governance

Automate martingale, parity, convexity, support, concentration, cross-engine, and rolling-stability checks.

17. Implementation pseudocode

17.1 Empirical CGF

```
function empirical_cgf(u, x, p):
    a_i = log(p_i) + u * x_i
    m = max(real-stabilizing component of a_i)
    return m + log(sum(exp(a_i - m)))
```

For complex u , use a real shift such as

$$m = \max_i \operatorname{Re}(\log p_i + ux_i)$$

so that the complex phase is retained.

17.2 Empirical Esscher calibration

```
function solve_esscher(carry, x, p):
    define g(theta):
        return cgf(theta + 1, x, p) - cgf(theta, x, p) - carry

    verify min(x) < carry < max(x)
    solve g(theta) = 0 by a bracketed root solver

    log_q_i = log(p_i) + theta * x_i
    normalize log_q with log-sum-exp
    return theta, q
```

17.3 Entropy projection

```
minimize sum_i q_i * log(q_i / p_i)
subject to:
    q_i >= 0
    sum_i q_i = 1
    sum_i q_i * exp(x_i) = forward_growth
    option bid <= sum_i q_i * discounted_payoff_i <= option_ask
```

17.4 Regime transform propagation

```

for each transform argument u:
    D[u] = diag(mgf_state_1(u), ..., mgf_state_K(u))
    M_h[u] = alpha_Q_transpose * matrix_power(P_Q * D[u], h) * ones

```

The $P^Q D(u)$ ordering assumes arriving-state emissions.

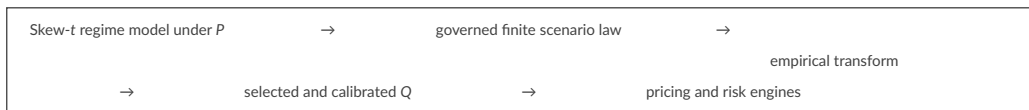
18. Research extensions

1. Time-varying transition probabilities driven by volatility, momentum, liquidity, and macro variables.
 2. State-dependent risk-neutral transition premia.
 3. Hierarchical shrinkage of λ_k toward zero and $1/\nu_k$ toward zero.
 4. Multivariate skew- t regimes with empirical copula or factor nodes.
 5. Martingale optimal transport between maturity-specific empirical marginals.
 6. Smooth entropy-regularized terminal densities rather than atomic laws.
 7. Joint calibration of returns, options, and variance swaps.
 8. Robust optimization over an ambiguity set around physical and risk-neutral scenario weights.
 9. Adjoint or automatic differentiation for calibration-aware Greeks.
 10. Regime-conditioned local or stochastic volatility overlays.
-

19. Final assessment

A mixed skew- t regime model can support coherent trading analytics and a derivative-pricing research platform if the physical and risk-neutral layers are kept conceptually separate.

The correct chain is



The finite scenario law is the primitive pricing model. Its empirical MGF is computationally powerful because it is finite and exact for that law. Its use is legitimate provided the model documentation states that finiteness comes from finite support. The framework does not restore a nonexistent population MGF for an unbounded Student- t .

The most defensible production choice is an option-constrained minimum-relative-entropy calibration on a governed finite scenario grid, with the forward imposed exactly, liquid option constraints applied within bid-ask, transition and tail risk premia regularized, and all prices cross-checked by direct summation, Fourier methods, and Monte Carlo.

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Source-verification note

Key source claims were checked against the original or institutional versions of Hansen's skew- t paper, Hamilton's regime-switching paper, Gerber and Shiu's Esscher-transform paper, Carr and Madan's FFT paper, and Breeden and Litzenberger's state-price-density result. The mathematical derivations in this study were independently rechecked for transform ordering, martingale conditioning, feasibility, convexity, and entropy-projection form.

20. Version 2 design principles and model boundary

20.1 Primitive object and transform hierarchy

The primitive one-step object in regime k is the governed scenario law

$$\mathcal{L}_k^P = \{(x_{k,i}, p_{k,i}^P)\}_{i=1}^{N_k}.$$

The empirical MGF, CGF, and characteristic function are derived representations:

$$\widehat{M}_k^P(u) = \sum_i p_{k,i}^P e^{ux_{k,i}}, \quad \widehat{\kappa}_k^P(u) = \log \widehat{M}_k^P(u), \quad \widehat{\phi}_k^P(v) = \widehat{M}_k^P(iv).$$

This hierarchy prevents an important category error. The framework is valid because it explicitly adopts a finite-state model. It is not valid because an empirical transform somehow regularizes the population transform of an unbounded skew- t law.

20.2 Three distinct probability objects

Production documentation and code must distinguish:

1. **Continuous physical reference law** F_k^P , used for estimation, interpretation, and generation of scenario candidates.
2. **Finite physical prior law** $\widehat{F}_k^P$, used as the governed scenario prior.
3. **Finite risk-neutral pricing law** $\widehat{F}_k^Q$, obtained by a stated measure-selection and calibration procedure.

The map

$$F_k^P \longrightarrow \widehat{F}_k^P \longrightarrow \widehat{F}_k^Q$$

contains two modelling interventions. The first is discretization and tail policy. The second is risk-neutralization. Their errors and sensitivities must be reported separately.

20.3 Interpretation of the calibrated model

When liquid options materially determine Q , the result should be described as an **option-implied finite state-price law regularized toward a skew- t regime prior**. It should not be presented as a pure skew- t option-pricing model. This wording correctly recognizes that the physical skew- t model may be only a prior once market option constraints are applied.

20.4 Intended scope

The architecture is most defensible for:

- physical density and regime forecasting;
- scenario-based risk measurement;
- European and discretely monitored path-dependent pricing;
- extraction of option-implied state prices;
- controlled comparisons between physical and risk-neutral probabilities;
- model-risk studies over alternative support and tail policies.

Additional modelling is required for deep-tail extrapolation, very long maturities, forward-skew products, continuously monitored barriers, American exercise under partial regime observation, and products whose value depends strongly on dynamics between calibrated maturities.

21. Statistical identification and physical calibration

21.1 Likelihood construction

Let $x_{1:T}$ denote observed log returns. For a K -state HMM with arriving-state emissions,

$$\ell(\Theta^P) = \sum_{t=1}^T \log \left[\sum_{j=1}^K \alpha_{t|t-1,j}^P f_j^P(x_t; \vartheta_j^P) \right],$$

where

$$\alpha_{t|t-1,j}^P = \sum_{i=1}^K \alpha_{t-1,i}^P p_{ij}^P,$$

and $\vartheta_j^P = (\mu_j^P, \sigma_j^P, \nu_j^P, \lambda_j^P)$. Evaluation must use scaled forward probabilities or log-domain recursions to avoid underflow.

21.2 Identification risks

Observed tail asymmetry can be explained by several substitutable mechanisms:

- lower degrees of freedom ν_k ;
- larger scale σ_k ;
- nonzero skew λ_k ;
- greater stress-state occupancy;
- more persistent stress transitions;
- or an additional jump or stress node.

Consequently, unconstrained estimation can produce apparently excellent in-sample fit with unstable live state probabilities. A disciplined nested sequence is preferred:

1. one-state Gaussian benchmark;
2. Gaussian HMM;
3. symmetric Student- t HMM;
4. selectively skewed Student- t HMM;
5. additional states only after out-of-sample evidence.

21.3 Penalized maximum likelihood

A representative physical objective is

$$\min_{\Theta^P} -\ell(\Theta^P) + \lambda_\lambda \sum_k \lambda_k^2 + \lambda_\nu \sum_k (\log \nu_k - \log \bar{\nu})^2 + \lambda_P \|P^P - \bar{P}\|_F^2.$$

The penalties shrink skewness toward symmetry, degrees of freedom toward a shared level, and transitions toward a stable reference matrix. Penalty strengths must be selected using rolling validation rather than in-sample likelihood alone.

21.4 Bayesian alternative

A Bayesian implementation may use:

- logistic-normal or Dirichlet priors for rows of P^P ;
- ordered priors on σ_k to reduce label switching;
- weakly informative priors on μ_k ;
- bounded transformed priors for λ_k ;
- priors on $\nu_k - 2$ or $\nu_k - 4$, depending on required moments.

Posterior uncertainty should propagate into scenario weights and pricing-risk bands. A single maximum-likelihood estimate understates model uncertainty, particularly for rare regimes.

21.5 State ordering and label control

A stable labeling convention may impose

$$\sigma_1 < \sigma_2 < \cdots < \sigma_K,$$

with secondary labels based on μ_k or λ_k . Ordering by volatility is operationally clearer than relabeling states after every calibration. If states overlap strongly, that is evidence that the selected K is too large.

21.6 Initialization and multi-start optimization

Recommended initialization:

1. cluster returns using robust scale and sign-sensitive features;
2. initialize μ_k and σ_k from trimmed cluster statistics;
3. initialize P^P with high diagonal persistence;
4. start with common ν and $\lambda = 0$;
5. release parameters gradually;
6. run multiple dispersed starts;
7. retain solutions only if filtered-state paths and parameters are stable.

A best likelihood from a single start is not sufficient evidence of a valid optimum.

21.7 Physical calibration diagnostics

At minimum, report:

- in-sample and rolling out-of-sample log scores;
- probability integral transform histograms;
- left- and right-tail exceedance rates;
- regime occupancy and expected duration;
- transition-count effective sample sizes;
- parameter standard errors or posterior intervals;
- filtered versus smoothed state paths;
- likelihood-profile plots for ν_k and λ_k ;
- performance against simpler nested models.

The skew parameter should be retained only if it improves genuine predictive or economic criteria.

22. Scenario construction and tail completion

22.1 Probability-preserving quantization

Partition the continuous reference law into cells $A_{k,i}$ and set

$$p_{k,i}^P = P_P(X \in A_{k,i} \mid S = k),$$

with representative node

$$x_{k,i} = E_P[X \mid X \in A_{k,i}, S = k]$$

when this conditional mean is numerically available. This preserves cell mass and generally reduces weak-integration error relative to arbitrary point selection.

22.2 Moment and payoff-aware quantization

If the scenario grid is intended for option pricing, approximation quality should not be judged only by ordinary moments. It should also control errors in economically relevant features:

$$1, \quad x, \quad x^2, \quad e^x, \quad (S_0 e^x - K_m)^+,$$

for representative strikes K_m . A grid that approximates the return density well may still approximate convex payoffs poorly.

22.3 Hybrid body, tail, and stress grid

A robust grid may combine:

- body quantile nodes from fitted skew- t regimes;
- historical crisis observations;
- option-implied right- and left-tail nodes;
- governed stress nodes with separately documented probabilities.

Each node must carry metadata identifying its origin. This permits attribution of prices and risks to historical, parametric, option-implied, and judgmental components.

22.4 Repeated-stress problem

If a daily stress endpoint can recur independently at every step, a long-horizon path may contain several implausibly repeated crises. The model should therefore support one or more of:

- a stress-arrival state with explicit intensity;
- a semi-Markov duration model;
- a cooldown state following a stress event;
- non-repeatable event flags over a specified horizon;
- horizon-specific terminal tail nodes;
- state-dependent stress-node probabilities.

This is important whenever stress nodes are not ordinary recurrent daily returns.

22.5 Support policy report

Every calibration should store:

- minimum and maximum node by regime and maturity;
- tail probability represented by each endpoint cell;
- whether the support is historical, truncated-parametric, spliced, or option-implied;
- all added stress nodes and their rationale;
- sensitivity under endpoint expansion and contraction;
- contribution of endpoint nodes to $E_Q[e^X]$ and option prices.

23. Risk-neutral calibration in detail

23.1 Calibration hierarchy

Use the following hierarchy rather than jumping directly to an unrestricted option fit:

1. verify support feasibility for the forward;
2. obtain a forward-consistent baseline Q ;
3. fit liquid European options within bid-ask;

- regularize state and transition risk premia;
- enforce cross-strike no-arbitrage;
- enforce or test cross-maturity consistency;
- validate out of sample and under support perturbations.

23.2 Forward-only minimum-KL projection

Given prior weights $p_i > 0$, solve

$$\min_q \sum_i q_i \log \frac{q_i}{p_i}$$

subject to

$$\sum_i q_i = 1, \qquad \sum_i q_i e^{x_i} = e^c.$$

The dual is one-dimensional. With convention

$$q_i(\beta) = \frac{p_i e^{\beta e^{x_i}}}{Z(\beta)}, \qquad Z(\beta) = \sum_j p_j e^{\beta e^{x_j}},$$

the root equation is

$$h(\beta) = \sum_i q_i(\beta) e^{x_i} - e^c = 0.$$

Its derivative is

$$h'(\beta) = \text{Var}_{q(\beta)}(e^X) \geq 0,$$

and is strictly positive for nondegenerate gross-return nodes. Therefore an interior feasible target has a unique finite β .

23.3 Multi-instrument entropy calibration

Let $G_{m,i}$ be the discounted payoff of instrument m at node i . With equality targets V_m , solve

$$\min_q D_{\text{KL}}(q||p)$$

subject to

$$Aq = b, \qquad q \geq 0.$$

The dual parameterization is

\$\$

$q_i(\boldsymbol{\beta})$

$$\frac{p_i \exp(\boldsymbol{\beta}^\top \mathbf{g}_i)}{\sum_j p_j \exp(\boldsymbol{\beta}^\top \mathbf{g}_j)}.$$

\$\$

The dual Hessian is the covariance matrix of the feature vector under q :

$$\nabla^2 \log Z(\boldsymbol{\beta}) = \text{Cov}_q(\mathbf{g}).$$

Near linear dependence among payoff features creates ill-conditioning. Strike selection, feature scaling, and regularization are therefore essential.

23.4 Bid-ask calibration

Exact fit to every midpoint can overfit stale or inconsistent quotes. Prefer

$$V_m^{\text{bid}} - \epsilon_m^- \leq \sum_i q_i G_{m,i} \leq V_m^{\text{ask}} + \epsilon_m^+,$$

with slack penalties

$$\lambda_s \sum_m [(\epsilon_m^-)^2 + (\epsilon_m^+)^2].$$

The forward, normalization, and nonnegativity constraints should remain exact unless data quality makes even the quoted forward unreliable.

23.5 Joint calibration of transitions and emissions

For a dynamic K -state model, variables may include

$$P^Q, \quad q_{k,i}^Q, \quad \alpha_0^Q.$$

A representative objective is

$$\begin{aligned} \mathcal{J} = & \sum_m w_m \rho \left(\frac{V_m^{\text{model}} - V_m^{\text{mid}}}{s_m} \right) + \lambda_q \sum_k D_{\text{KL}}(q_k^Q \| p_k^P) \\ & + \lambda_P \sum_i D_{\text{KL}}(p_i^Q \| p_i^P) + \lambda_T \mathcal{R}_T + \lambda_C \mathcal{R}_C, \end{aligned}$$

where ρ is a robust loss, $\mathcal{R}_T$ penalizes maturity instability, and $\mathcal{R}_C$ penalizes concentration. Constraints include

$$\begin{aligned} P^Q \mathbf{1} &= \mathbf{1}, & P^Q &\geq 0, \\ \sum_i q_{k,i}^Q &= 1, & q_{k,i}^Q &\geq 0, \end{aligned}$$

and

$$P^Q \mathbf{m}^Q(1) = e^c \mathbf{1}.$$

23.6 Identifiability under Q

Vanilla options typically identify terminal state-price distributions more strongly than they identify a unique decomposition into transition and emission premia. Therefore:

- do not freely calibrate all entries of P^Q and all node weights without shrinkage;
- use economically interpretable low-dimensional transition tilts;
- examine the profile objective along weakly identified directions;
- report multiple near-optimal calibrations;
- use exotic or forward-start information only when liquid enough to add identification.

23.7 Transition-matrix parameterizations

Useful parameterizations include:

Row-wise exponential tilt

$$p_{ij}^Q = \frac{p_{ij}^P \exp(\eta_{ij})}{\sum_{\ell} p_{i\ell}^P \exp(\eta_{i\ell})}.$$

Low-rank risk-premium tilt

$$\eta_{ij} = a_i + b_j + \mathbf{z}_i^\top R \mathbf{z}_j.$$

Continuous-time generator

Estimate a valid generator G^Q and set

$$P^Q(\Delta) = \exp(\Delta G^Q).$$

The generator approach gives time-step consistency but requires nonnegative off-diagonal intensities and zero row sums.

23.8 Calibration loss conventions

Possible quote losses include price error, implied-volatility error, and vega-scaled price error. Vega scaling is often useful:

$$\varepsilon_m^{\text{vega}} = \frac{V_m^{\text{model}} - V_m^{\text{market}}}{\max(\text{Vega}_m, \text{Vega}_{\min})}.$$

However, deep options with poor liquidity should not gain excessive influence merely because of the scaling convention. Use liquidity weights, bid-ask widths, robust losses, and explicit quote filters.

23.9 Calibration algorithm

A stable sequence is:

1. clean quotes and infer discount factors and forwards;
2. enforce or project each market slice onto static-arbitrage constraints;
3. build a support wide enough for all retained strikes;
4. solve the forward-only entropy projection;
5. add option constraints from liquid to less liquid strikes;
6. calibrate low-dimensional transition premia;
7. expand degrees of freedom only if validation improves;
8. calculate exact constraint residuals;
9. perturb quotes, support, priors, and starting values;
10. retain calibration ensembles when non-uniqueness is material.

23.10 Posterior-prior displacement diagnostics

Report:

$$D_{\text{KL}}(Q\|P), \quad D_{\text{KL}}(P\|Q), \quad \frac{1}{2} \sum_i |q_i - p_i|,$$

as well as Wasserstein distance, maximum likelihood ratio $\max_i (q_i/p_i)$, minimum likelihood ratio, tail-mass migration, and effective scenario count. These reveal whether the physical skew- t prior still materially influences the calibrated pricing law.

24. Maturity consistency and dynamic completion

24.1 Marginal versus dynamic calibration

A separate terminal law Q_T for each maturity may price European options correctly while failing to define a common process. Dynamic products require transition kernels K_{T_a, T_b} satisfying

$$q_{T_b}(y) = \sum_x q_{T_a}(x) K_{T_a, T_b}(x, y)$$

and the martingale restriction

$$\sum_y K_{T_a, T_b}(x, y) S(y) = S(x) e^{c_{a,b}}.$$

24.2 Dynamic completion alternatives

Possible approaches are:

- calibrate one common regime process jointly to all maturities;
- construct martingale transport couplings between calibrated marginals;
- fit maturity-dependent but smoothly varying transition matrices;
- use an entropy-minimizing path measure subject to all option constraints;
- use a Markov-chain projection that matches marginal state prices.

24.3 Calendar-arbitrage checks

For deterministic discounting, call prices expressed as functions of forward moneyness should satisfy suitable calendar monotonicity conditions. At a minimum:

- total implied variance should not create calendar-spread arbitrage;
- later marginal laws must be reachable from earlier laws under a martingale coupling;
- interpolated maturities must preserve convexity and forward consistency.

24.4 Path-space entropy calibration

Let ω denote a complete regime and return-node path with physical prior probability $P(\omega)$. A dynamic minimum-entropy model solves

$$\min_Q \sum_{\omega} Q(\omega) \log \frac{Q(\omega)}{P(\omega)}$$

subject to normalization, conditional martingale restrictions, and option-price constraints across maturities. This is conceptually clean but computationally large. Dynamic programming, factorized likelihood ratios, iterative proportional fitting, or stochastic optimization may be required.

25. Analytical and semi-analytical pricing

25.1 Exact finite-sum formula

For terminal nodes (x_i, q_i) and payoff Π ,

$$V_0 = D(0, T) \sum_i q_i \Pi(S_0 e^{x_i}).$$

This is the primary analytical formula under a terminal finite law. For European calls,

$$C_0 = D(0, T) \sum_i q_i (S_0 e^{x_i} - K)^+.$$

For a discrete law, a strip of strikes can be priced efficiently after sorting terminal prices and precomputing right-tail sums:

$$A_j = \sum_{i \geq j} q_i S_i, \quad B_j = \sum_{i \geq j} q_i.$$

Then for a strike between nodes $S_{j-1} < K \leq S_j$,

$$C(K) = D(0, T)(A_j - K B_j).$$

This gives near-constant-time evaluation per strike after sorting.

25.2 Matrix-transform formula

For homogeneous transitions and arriving-state emissions,

$$M_h^Q(u) = \alpha_0^{Q^\top} [P^Q D^Q(u)]^h \mathbf{1}.$$

For time-varying dynamics,

$$M_{0,h}^Q(u) = \alpha_0^{Q^\top} \prod_{n=1}^h [P_n^Q D_n^Q(u)] \mathbf{1},$$

with chronological matrix ordering fixed explicitly.

25.3 Closed forms under special emission choices

True closed forms may exist if regime-conditional dynamics are Gaussian, affine, phase-type, or otherwise transform-tractable. The general empirical atomic model does not require a continuous-density closed form because direct summation is already exact. Analytical approximations are mainly useful for:

- controls in Monte Carlo;
- initial guesses in calibration;
- sanity checks;
- fast approximations when the terminal support is extremely large.

25.4 Moment-matched approximations

A Gaussian, shifted-lognormal, or saddlepoint approximation can be fitted to selected cumulants of the finite law. Such approximations must not replace the exact finite-sum benchmark without documented error. Saddlepoint formulas also require care near support boundaries and payoff discontinuities.

26. Deterministic quadrature

26.1 When quadrature is appropriate

For the unsmoothed atomic law, quadrature is unnecessary because the expectation is a finite sum. Quadrature becomes useful when:

- continuous skew- t bodies are truncated and retained;
- smooth tail completion is attached;
- kernel smoothing replaces each atom by a local density;
- a conditional continuous factor remains after discrete regime conditioning;
- calibration integrates over latent variables.

26.2 Domain decomposition

For a European call, split at

$$x_K = \log(K/S_0).$$

Then

$$C_0 = D(0, T) \int_{x_K}^U (S_0 e^x - K) f_T^Q(x) dx,$$

plus any explicit tail contribution beyond U . Splitting at payoff kinks prevents an adaptive integrator from wasting evaluations around nonsmooth points.

26.3 Recommended rules

- **Gauss-Kronrod:** general one-dimensional adaptive benchmark.
- **Gauss-Legendre:** efficient after mapping to a finite interval.
- **Clenshaw-Curtis:** useful for nested grids and repeated evaluations.
- **Double-exponential or tanh-sinh:** useful near endpoint singularities or difficult transformed tails.
- **Gauss-Hermite:** useful for conditionally Gaussian factors, not as a default rule for skew- t tails.
- **Sparse grids:** possible for a small number of continuous factors, but dimensionality grows rapidly.

26.4 Error controls

Separate:

\$\$

\text{total error}

= \text{tail truncation error}

- \text{quadrature error}
 - \text{interpolation error}
 - \text{model error}.
- \$\$

Run absolute and relative tolerance tests, interval expansion tests, and independent comparison with a high-order or arbitrary-precision benchmark for selected cases.

27. Fourier quadrature, FFT, and COS methods

27.1 Characteristic function

For a terminal finite law,

$$\phi_T(v) = \sum_i q_i e^{ivx_i}.$$

For regime dynamics it follows from the matrix product. Complex arguments required by damped pricing must be evaluated using real-part stabilization.

27.2 Direct Fourier quadrature

For isolated strikes, direct adaptive integration in Fourier space is often preferable to FFT. It avoids grid-induced strike interpolation and allows local control of oscillatory error. The implementation must specify:

- transform convention;
- damping or contour shift;
- integration cutoff;
- oscillatory quadrature rule;
- handling near zero frequency;
- branch conventions for complex logarithms.

27.3 Carr-Madan FFT

For log strike k , apply damping

$$\tilde{C}(k) = e^{\alpha k} C(k), \quad \alpha > 0,$$

and use the shifted characteristic function. Under finite support all exponential moments exist, but large α can still cause severe scaling and cancellation. Numerical admissibility is not the same as mathematical existence.

A practical FFT implementation must manage:

- frequency spacing η ;
- number of points N ;
- log-strike spacing $\lambda = 2\pi/(N\eta)$;
- strike range;
- Simpson or higher-order weights;
- phase shifts;
- aliasing;
- interpolation back to quoted strikes.

27.4 COS method

The COS method expands the density or payoff over a finite interval $[a, b]$. It can be highly efficient for smooth distributions with well-controlled truncation. For an atomic law, the characteristic function is easy to evaluate, but Gibbs-type behavior and nonsmooth Greeks can arise because the terminal measure contains atoms. Smoothing or direct summation may therefore be superior unless multi-step propagation has produced a sufficiently regular grid law.

27.5 Fourier-space time stepping

For Bermudan, barrier, and regime-switching problems, one may alternate:

1. transform the state-conditioned value function;
2. multiply by the one-step transition-emission symbol;
3. invert the transform;
4. apply barriers or early-exercise conditions;
5. repeat backward in time.

This is attractive when transition is convolutional in log price. Care is required around barriers, absorbing states, wraparound, and discontinuous exercise boundaries.

27.6 Fourier validation

Require convergence under simultaneous changes in:

- N ;
- frequency spacing;
- damping;

- log-strike domain;
- truncation interval;
- interpolation scheme.

A single apparently stable FFT price is not evidence of convergence. Compare each test case against direct summation or deterministic convolution.

28. Monte Carlo and quasi-Monte Carlo

28.1 Exact simulation of the finite model

No Euler discretization is required for the one-step finite scenario process. Given state $S_t = i$:

1. sample arriving state $S_{t+1} = j$ from row i of P^Q ;
2. sample return node $x_{j,n}$ from $q_{j,n}^Q$;
3. update $S_{t+1}^{\text{asset}} = S_t^{\text{asset}} e^{x_{j,n}}$;
4. update path statistics and continue.

The only time discretization is the model's own step size.

28.2 Variance reduction

Useful methods include:

- analytical or direct-sum European control variates;
- stratification by current state or terminal state;
- stratification by number of stress-state visits;
- conditional Monte Carlo over return nodes given the regime path;
- common random numbers across calibration candidates;
- randomized quasi-Monte Carlo for inverse-CDF draws;
- importance sampling toward barrier or tail events.

Antithetic sampling is not automatically effective for asymmetric discrete laws and should be validated rather than assumed.

28.3 Conditional Monte Carlo

If the regime path is simulated but emission aggregation can be convolved exactly conditional on that path, integrate over emissions rather than sampling them. Conversely, if the number of regime paths is manageable, sum over regime paths and simulate only continuous overlays. Rao-Blackwellization can materially reduce variance.

28.4 Rare-event importance sampling

For a rare event A , simulate under proposal $\tilde{Q}$ and use

$$V_0 = D(0, T) E_{\tilde{Q}} \left[\Pi \frac{dQ}{d\tilde{Q}} \right].$$

In a finite state-path model, the likelihood ratio is the product of transition and emission probability ratios. Proposal probabilities must remain positive wherever the original model assigns positive probability to a payoff-relevant path.

28.5 Greeks

Methods include:

- bump-and-revalue with common random numbers;
- likelihood-ratio estimators for probability parameters;
- pathwise derivatives when payoff and parameterization are smooth;
- algorithmic differentiation through calibration and simulation;
- conditional smoothing around payoff kinks.

For atomic laws, naive pathwise gamma is unreliable because exercise indicators jump at node crossings.

28.6 Monte Carlo stopping rule

Report the discounted standard error

$$\widehat{\text{SE}} = D(0, T) \frac{s_{\Pi}}{\sqrt{M}},$$

confidence intervals, effective sample size under importance sampling, and replication-to-replication stability. Stop only when both statistical error and implementation bias are below product-specific tolerances.

29. Trees, lattices, and dynamic programming

29.1 Nonrecombining scenario tree

A literal tree branches by arriving regime and return node. Its node count grows exponentially. It is exact for short horizons but quickly becomes infeasible.

29.2 Recombining log-price lattice

Map log prices onto grid points $y_m = y_0 + m\Delta y$. After each transition, distribute probability to neighboring grid points while preserving, where possible:

$$\sum_m w_m = 1, \quad \sum_m w_m e^{y_m} = \text{target gross first moment}.$$

Moment-preserving interpolation is preferable to nearest-grid projection. Record mass and martingale residuals after every step.

29.3 State-conditioned Bellman recursion

For regime-observable value $V_n(s, k)$,

$$V_n(s, k) = D_n \sum_j p_{kj}^Q \sum_i q_{j,i}^Q V_{n+1}(se^{x_{j,i}}, j).$$

For a Bermudan option,

$$V_n(s, k) = \max \left\{ \Pi_n(s), D_n \sum_j p_{kj}^Q \sum_i q_{j,i}^Q V_{n+1}(se^{x_{j,i}}, j) \right\}.$$

29.4 Partial observation

If the regime is unobservable, the sufficient state is generally the belief vector

$$\alpha_t = (P(S_t = 1 \mid \mathcal{F}_t), \dots, P(S_t = K \mid \mathcal{F}_t)).$$

An exact tree in (s, α) is high-dimensional. Approaches include belief-grid interpolation, particle filtering inside simulation, regression Monte Carlo, and approximations that freeze or compress the belief state. Pricing as if the latent state were directly observable can materially overstate the information available to the exercise policy.

29.5 Tree validation

Check:

- probability conservation;
- one-step and multi-step martingale residuals;
- convergence under Δy and Δt refinement;
- sensitivity to interpolation;
- agreement with direct sums for European claims;
- monotonicity and convexity in strike;
- stability of early-exercise boundaries.

30. Finite difference, PIDE, and Fourier-space time stepping

30.1 Important model distinction

A pure finite-scenario discrete-time process does not naturally generate a classical local diffusion PDE. Its backward equation is an expectation recursion. A finite-difference PDE becomes appropriate only after embedding or approximating the model by a continuous-time diffusion, jump process, or Markov-chain generator.

30.2 Continuous-time regime-switching generator

Suppose regime k has local diffusion coefficients μ_k^Q and σ_k and generator $G^Q = (g_{k\ell}^Q)$. Then a European value vector $V_k(t, s)$ satisfies a coupled system of the form

$$\frac{\partial V_k}{\partial t} + (r - q)s \frac{\partial V_k}{\partial s} + \frac{1}{2} \sigma_k^2 s^2 \frac{\partial^2 V_k}{\partial s^2} - rV_k + \sum_{\ell \neq k} g_{k\ell}^Q (V_\ell - V_k) = 0.$$

If jumps or a continuous tail-completion component are included, add the compensated integral operator, producing a PIDE.

30.3 Discrete scenario jump operator

For a continuous-time finite jump measure with intensities $\lambda_{k,i}$ and log jumps $x_{k,i}$, the jump term is

$$\mathcal{J}_k V(s) = \sum_i \lambda_{k,i} [V(se^{x_{k,i}}) - V(s) - s(e^{x_{k,i}} - 1)V_s(s)],$$

subject to the chosen truncation convention. Because the jump support is finite, the integral operator becomes a weighted sum but still requires interpolation at off-grid values $se^{x_{k,i}}$.

30.4 Schemes

Recommended schemes include:

- fully implicit Euler for robustness;

- Crank-Nicolson with initial damping steps for nonsmooth payoffs;
- IMEX schemes treating local diffusion implicitly and jump terms explicitly or semi-implicitly;
- operator splitting between diffusion, jumps, and regime coupling;
- penalty or policy-iteration methods for American exercise;
- monotone schemes where preservation of no-arbitrage properties is critical.

30.5 Stability and positivity

The numerical operator should preserve, as far as possible:

- nonnegative option values;
- monotonicity in spot;
- convexity for vanilla payoffs;
- appropriate boundary behavior;
- stable regime coupling.

Central differencing can create oscillations when drift dominates. Upwinding, flux limiting, or transformed coordinates may be necessary.

30.6 Boundary conditions

For a call on a non-dividend-paying asset, typical asymptotic boundaries are

$$V(t, 0) = 0,$$

and at large $S_{\max}$,

$$V(t, S_{\max}) \approx S_{\max} - KD(t, T),$$

adjusted for dividends and product features. Barrier and rebate contracts require contract-specific boundary conditions. Boundary truncation must be tested separately from spatial and temporal discretization.

30.7 American complementarity problem

For each regime,

$$\max (\mathcal{L}^Q V - rV + V_t, \Pi - V) = 0,$$

with coupling between regimes. Penalty methods, projected iterative solvers, or policy iteration can solve the resulting discrete linear-complementarity system.

30.8 When FD or PIDE is worth using

FD/PIDE methods are attractive for:

- American and Bermudan claims;
- barriers and rebates;
- state-dependent boundaries;
- models with local diffusion overlays;
- products naturally expressed by backward state-space recursion.

They are usually unnecessary for a one-maturity European payoff under a modest atomic terminal law, where direct summation is exact and simpler.

31. Engine-selection matrix

Use the following practical hierarchy:

- **European, modest terminal support:** direct finite sum.
- **Many strikes, large propagated support:** sorted tail sums, FFT, or COS after validation.
- **One or a few strikes with smooth continuous completion:** adaptive quadrature or direct Fourier quadrature.
- **Path-dependent, high-dimensional, or exposure simulation:** Monte Carlo or quasi-Monte Carlo.
- **Bermudan or American with manageable state dimension:** lattice, dynamic programming, FD, or LSMC.
- **Barriers with convolutional transitions:** lattice, Fourier-space time stepping, FD/PIDE, or Brownian/jump bridge-adjusted MC where applicable.
- **Partial observation:** simulation or belief-state dynamic programming.
- **Calibration loop:** use the fastest engine that remains exact or tightly controlled for the quote set, with slower engines as independent validators.

No single engine should be declared universally best. Product structure, state dimension, monitoring convention, calibration frequency, and required Greeks determine the choice.

32. Software architecture and implementation specification

32.1 Core modules

A production library should separate:

1. `physical_model` : skew- t density, HMM filtering, estimation, diagnostics.
2. `scenario_builder` : quantization, historical nodes, stress nodes, tail completion.
3. `transform_engine` : MGF, CGF, CF, matrix products, stabilization.
4. `risk_neutral_calibrator` : Esscher, entropy, bid-ask, transition calibration.
5. `distribution_propagator` : convolution, lattice, transform, and path simulation.
6. `pricing_engines` : direct sum, quadrature, Fourier, FFT, COS, MC, tree, FD/PIDE.
7. `greeks_engine` : fixed-law, recalibrated, and adjoint sensitivities.
8. `validation` : martingale, arbitrage, convergence, support, and concentration tests.
9. `market_data` : curves, forwards, dividends, corporate actions, quote cleaning.
10. `audit_store` : inputs, parameters, solver traces, seeds, tolerances, and hashes.

32.2 Canonical data structures

A scenario law should store:

```
ScenarioLaw
  nodes_log_return[N]
  probabilities[N]
  regime_id
  horizon
  measure_tag          # P-prior or Q-pricing
  source_tag[N]        # historical, quantile, stress, implied
  support_policy
  tail_metadata
  calibration_id
```

A dynamic model should store transition conventions explicitly:

```

RegimeModel
    transition_matrix[K,K]
    emission_laws[K]
    emission_timing      # arriving-state or departing-state
    time_step
    initial_state_prob[K]
    rate_curve_id
    dividend_curve_id

```

32.3 Numerical safeguards

Mandatory safeguards include:

- log-sum-exp normalization;
- scaled HMM forward recursion;
- bracketed scalar root solvers;
- constrained convex solvers for entropy calibration;
- row-stochasticity projections only with audit logging;
- deterministic random seeds for regression tests;
- explicit complex transform conventions;
- checks for NaN, infinity, negative probabilities, and lost mass;
- compensated or pairwise summation where weight scales are extreme.

32.4 Pseudocode for robust entropy calibration

```

function calibrate_entropy(prior_p, features_G, targets_or_bounds):
    validate prior_p > 0 and sum(prior_p) = 1
    scale each feature column
    test convex-hull feasibility or solve a feasibility program
    initialize dual variables at zero
    solve convex dual using damped Newton or trust region
    recover q in log space
    verify q >= 0 and sum(q) = 1
    verify forward residual at strict tolerance
    verify option residuals or bid-ask inclusion
    compute Hessian condition number and displacement diagnostics
    return q, dual_variables, diagnostics

```

32.5 Pseudocode for exact regime simulation

```

function simulate_path(model_Q, spot0, steps, rng):
    state = draw(model_Q.initial_state_prob, rng)
    spot = spot0
    path = [spot]
    for n in 1..steps:
        next_state = draw(model_Q.P_Q[state, :], rng)
        x = draw(model_Q.emission_law_Q[next_state], rng)
        spot = spot * exp(x)
        state = next_state
        append(path, spot)
    return path

```

32.6 Reproducibility

Every result must retain:

- source-data timestamp and checksum;
- code version and build identifier;

- model configuration;
- optimizer, tolerances, and stopping reason;
- random-number generator and seed;
- support and tail policy;
- excluded quotes and reasons;
- calibration residuals;
- validation results;
- hardware and precision where numerically material.

33. Greeks, hedging, and calibration risk

33.1 Three Greek definitions

Distinguish:

1. **Frozen-law Greek:** nodes and Q weights fixed.
2. **Market-rule Greek:** curves, forward, and smile moved according to a stated sticky convention.
3. **Recalibrated Greek:** the complete model is recalibrated after the market bump.

These can differ materially. A risk report must identify which definition is used.

33.2 Implicit differentiation through calibration

If calibrated parameters $\theta^*(z)$ solve

$$F(\theta, z) = 0,$$

then, when regularity holds,

$$\frac{d\theta^*}{dz} = - \left(\frac{\partial F}{\partial \theta} \right)^{-1} \frac{\partial F}{\partial z}.$$

For value $V(\theta^*(z), z)$,

$$\frac{dV}{dz} = V_z + V_\theta \frac{d\theta^*}{dz}.$$

This is preferable to noisy repeated optimization when the active constraint set is stable. Bid-ask inequalities and changing active sets can create nonsmooth sensitivities.

33.3 Model-risk Greeks

In addition to market Greeks, calculate sensitivity to:

- support endpoints;
- stress-node probabilities;
- KL regularization strength;
- transition-risk-premium penalty;
- number of regimes;
- degrees-of-freedom constraints;
- maturity-coupling strength;
- smoothing bandwidth.

These are not hedge ratios, but they quantify valuation uncertainty caused by modelling choices.

34. Validation, benchmarking, and acceptance criteria

34.1 Model benchmarks

Compare against:

- Black-Scholes or Bachelier where appropriate;
- Gaussian mixture;
- symmetric Student- t regime model;
- two-state versus three-state HMM;
- direct option-implied discrete density without a skew- t prior;
- a standard stochastic-volatility or exponential-Lévy benchmark where relevant.

Complexity is justified only if it improves out-of-sample performance, stability, or product coverage.

34.2 Synthetic recovery tests

Generate data from known parameter sets and verify:

- parameter recovery under adequate samples;
- regime classification accuracy;
- transition recovery;
- calibration recovery from synthetic option prices;
- robustness when the fitted model is misspecified;
- confidence-interval coverage.

34.3 Engine unit tests

Include cases with:

- one node, where results are deterministic;
- two nodes with hand-calculable prices;
- one regime reducing to an ordinary finite law;
- identical regime emissions, making transitions irrelevant to terminal returns;
- zero rates and dividends;
- strikes below all nodes and above all nodes;
- limiting probabilities approaching zero;
- boundary forward feasibility.

34.4 Cross-engine acceptance

For a benchmark price V_* :

$$|V_{\text{quad}} - V_*| \leq \varepsilon_{\text{quad}},$$

$$|V_{\text{FFT}} - V_*| \leq \varepsilon_{\text{FFT}},$$

and

$$|V_{\text{MC}} - V_*| \leq z_{1-\alpha/2} \widehat{\text{SE}}_{\text{MC}} + \varepsilon_{\text{bias}}.$$

FD and lattice errors should exhibit the expected refinement behavior until interpolation, boundary, or payoff nonsmoothness dominates.

34.5 Calibration acceptance

A calibration is accepted only if:

- normalization and nonnegativity pass strict tolerances;
- the forward martingale residual passes;
- retained liquid options lie within bid-ask or approved tolerances;
- strike monotonicity and convexity pass;
- calendar checks pass;
- weight concentration remains within limits;
- support perturbations do not cause unacceptable price changes;
- multiple starts reach economically consistent solutions;
- calibration displacement from the prior is explainable.

34.6 Challenger and fallback

Production governance should define:

- a simpler challenger model;
 - conditions triggering fallback;
 - stale-market procedures;
 - maximum permitted calibration displacement;
 - support-extension rules;
 - limits on stress-node contribution;
 - manual override and approval requirements.
-

35. Revised implementation roadmap

Phase 1: deterministic one-period prototype

Implement scenario laws, stable transforms, direct pricing, Esscher calibration, forward-only entropy projection, and exact unit tests.

Phase 2: physical HMM

Implement Gaussian and symmetric- t benchmarks before selective skew- t regimes. Add rolling filtering, forecast scoring, and parameter-stability diagnostics.

Phase 3: option-constrained terminal calibration

Add quote cleaning, bid-ask entropy calibration, static-arbitrage checks, tail-node governance, and posterior-prior displacement reports.

Phase 4: multi-step dynamics

Implement regime-aware convolution and exact Monte Carlo. Validate all European claims against deterministic propagation.

Phase 5: Fourier engines

Add direct Fourier quadrature, FFT, and COS only after transform conventions and direct benchmarks are stable.

Phase 6: early exercise and barriers

Add recombining lattice or grid dynamic programming, then FD/PIDE or Fourier-space time stepping where a continuous-time embedding is justified. Add LSMC for higher-dimensional path dependence.

Phase 7: maturity consistency

Jointly calibrate a dynamic model or construct martingale couplings between maturity slices. Do not claim dynamic exotic capability before this layer passes validation.

Phase 8: production controls

Add calibration ensembles, audit persistence, performance engineering, parallel pricing, automatic differentiation, monitoring dashboards, and independent model validation.

36. Revised final assessment

The framework is best understood as a governed finite-state pricing and risk architecture with a skew- t regime model supplying physical structure and prior information. The empirical MGF is an exact and useful transform of the finite law, but it is not the source of economic legitimacy and does not restore the missing population MGF of an unbounded Student- t law.

The most defensible chain is

continuous skew- t HMM under $P \rightarrow$ governed finite P prior $\rightarrow$ martingale- and option-constrained finite Q
--

The preferred European calibration is an option-constrained minimum-relative-entropy projection with exact forward consistency, bid-ask treatment of quotes, explicit support and tail governance, and quantified displacement from the physical prior. For dynamic and path-dependent products, separately calibrated terminal laws are insufficient. A common risk-neutral process or a valid sequence of martingale transition kernels is required.

Direct finite summation should remain the principal benchmark. Quadrature is appropriate for continuous completion; Fourier quadrature, FFT, and COS are scale tools; Monte Carlo is the flexible path-dependent engine; trees and lattices support transparent dynamic programming; and FD/PIDE methods are justified only after a continuous-time diffusion or jump embedding has been specified. Engine choice must follow the product and model structure rather than fashion.

The decisive empirical question is whether this architecture improves out-of-sample density forecasts, option fits, stability, or product coverage relative to simpler models after complexity and model risk are penalized. If it does not, the simpler challenger should be preferred. If it does, the framework can serve as a coherent bridge between physical heavy-tail regime analysis, option-implied state prices, and controlled derivative-pricing computation.

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